

HONGBO WEN

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I am a Ph.D. candidate in Computer Science at UC Santa Barbara, advised by Yu Feng. My research develops program analysis and synthesis techniques to secure complex software systems, with applications in **LLM agent security**, **cryptography & zero-knowledge proofs**, and **AI-augmented vulnerability discovery**. I have discovered zero-day vulnerabilities across LLM agent platforms, DeFi protocols, and ZK circuits, and have advised 50+ engineers on shipping security-critical production systems and AI-agent native engineering transformation. My work has been funded by **Ethereum Foundation** and conducted in collaboration with **Google Security** and **a16z crypto**.

EDUCATION

University of California, Santa Barbara, CA, USA

Sep 2022 - Present

Ph.D. candidate in Computer Science (expected 2027)

GPA 4.0/4.0

Tsinghua University, Beijing, China

Aug 2015 - Jun 2020

Bachelor of Engineering

SELECTED PUBLICATIONS

- [1] Ying Li, Yanju Chen, **Hongbo Wen**, Bosi Zhang, Hanzhi Liu, Peiran Wang, Yu Feng, Yuan Tian. *VIGIL: Runtime Enforcement of Behavioral Specifications in AI Agent Skills*. **arXiv 2026**.
- [2] Ying Li, **Hongbo Wen**, Yanju Chen, Hanzhi Liu, Yuan Tian, Yu Feng. *No Attack Required: Semantic Fuzzing for Specification Violations in Agent Skills*. **arXiv 2026**.
- [3] **Hongbo Wen**, Ying Li, Hanzhi Liu, Chaofan Shou, Yanju Chen, Yuan Tian, Yu Feng. *SEMIA: Auditing Agent Skills via Constraint-Guided Representation Synthesis*. **arXiv 2026**.
- [4] Hanzhi Liu, Chaofan Shou, Xiaonan Liu, **Hongbo Wen**, Yanju Chen, Ryan Jingyang Fang, Yu Feng. *Synthesizing Multi-Agent Harnesses for Vulnerability Discovery*. **arXiv 2026**.
- [5] Hanzhi Liu, Chaofan Shou, **Hongbo Wen**, Yanju Chen, Ryan Jingyang Fang, Yu Feng. *Your Agent Is Mine: Measuring Malicious Intermediary Attacks on the LLM Supply Chain*. **arXiv 2026**.
- [6] Junrui Liu, Jiabin Song, Yanning Chen, Hanzhi Liu, **Hongbo Wen**, Luke Pearson, Yanju Chen, Yu Feng. *Tabby: A Synthesis-Aided Compiler for High-Performance Zero-Knowledge Proof Circuits*. **OOPSLA 2025**.
- [7] **Hongbo Wen**, Hanzhi Liu, Yanju Chen, Jingyu Ke, Dahlia Malkhi, Yu Feng. *Thunderbolt: Fast Asynchronous Off-Chain Bitcoin Transfers*. **IACR ePrint 2025**.
- [8] **Hongbo Wen**, Hanzhi Liu, Jiabin Song, Yanju Chen, Wenbo Guo, Yu Feng. *FORAY: Towards Effective Attack Synthesis against Deep Logical Vulnerabilities in DeFi Protocols*. **ACM CCS 2024**.
- [9] **Hongbo Wen**, Jon Stephens, Yanju Chen, Kostas Ferles, Shankara Pailoor, Kyle Charbonnet, Isil Dillig, Yu Feng. *Practical Security Analysis of Zero-Knowledge Proof Circuits*. **USENIX Security 2024**.

INVITED TALKS

Tabby: A Synthesis-Aided Compiler for High-Performance Zero-Knowledge Proof Circuits Feb 2026

SoCalPLS

USC, Los Angeles, CA, USA

Tabby: Automated Programming of Efficient Zero-Knowledge Proof Circuits Feb 2025

ETHDenver

Denver, CO, USA

Practical Security Analysis of Zero-Knowledge Proof Circuits <i>SoCalPLS</i>	Feb 2025 UCSD, San Diego, CA, USA
FORAY: Towards Effective Attack Synthesis against Deep Logical Vulnerabilities in DeFi Protocols <i>ACM CCS '24</i>	Oct 2024 Salt Lake City, UT, USA
Practical Security Analysis of Zero-Knowledge Proof Circuits <i>USENIX Security '24</i>	Aug 2024 Philadelphia, PA, USA

SELECTED GRANTS

Ethereum Foundation Academic Grants Round 2024 Yu Feng, Hanzhi Liu, <u>Hongbo Wen</u> <i>Topic: Sentinel: Adaptive Counter-Attack Synthesis for Mitigating Onchain Exploits</i>	Jun 2024
Ethereum Foundation Academic Grants Round 2023 Yu Feng, Yanju Chen, <u>Hongbo Wen</u> <i>Topic: Financial Model-Driven Attack Synthesis for DeFi</i>	Jun 2023

SELECTED ZERO-DAY DISCOVERIES

- [1] **LLM Agents:** Filed **50+** **zero-day** security advisories on OpenClaw/ClawHub agent skills, covering credential leakage, RCE, PII exfiltration, and indirect prompt injection.
- [2] **DeFi:** Discovered **10 zero-day** vulnerabilities on BNB Chain DeFi protocols using *FORAY*, with verified exploit traces matching $\geq$ \$21M historical-loss patterns.
- [3] **ZK Circuits:** Discovered zero-day vulnerabilities in widely-used Circom projects using *ZKAP*, including **0xPARC**, **Polygon-ZK**, **Axiom**, **iden3**, **privacy-ethereum**.

WORK EXPERIENCE

Riema Labs Inc. Technical Advisor. Highlights: - Advised a 50+ engineering team on shipping security-critical production systems and AI-agent native engineering transformation, serving 150K+ users. - Shipped cross-chain bridges, Bitcoin-native ZK infrastructure, and user-custody wallet systems.	Jan 2024 - May 2026
Veridise Inc. Research & Development Engineer: smart contract and zero-knowledge proof security. Highlights: Discovered 30+ critical vulnerabilities in widely-used ZK projects, and 10+ zero-day DeFi vulnerabilities.	Mar 2022 - Aug 2022

SERVICE

USENIX Security '25 Artifact Evaluation Committee	2025
CS162 Programming Languages , Teaching Assistant, UC Santa Barbara	2023 – 2025

SKILLSET

Languages: Rust, Go, Solidity, C/C++, JavaScript, Python.

AI-Assisted Research: Several projects and publications herein were co-developed with LLM coding agents, directed and verified by me.